Reducing Output Size

Case Study: Frequent Itemsets

(Part 2)

OUTLINE

1. Introduction to market-basket analysis (Part 1)
2. Algorithms for frequent itemset mining (Part 1)
    - Sequential algorithms: A-Priori.
    - Approaches to handle large inputs.
3. Controlling the output size (Part 2)
    - (Frequent) closed itemsets.
    - Maximal itemsets.
    - Top-K frequent (closed) itemsets.

Limitations of frequent patterns mining

1. Redundancy: many returned patterns may characterize the same subpopulation of data (e.g., customers).
2. Difficult control of output size: it is hard to predict how many patterns will be returned for a given support threshold.
3. Significance: are the returned patterns significant, interesting?

Will address the first two issues

Observation: *we focus on itemsets, however, many considerations apply to other patterns as well* (e.g., association rules).

(Frequent) closed and maximal itemsets

Let T be a dataset of N transactions over I, and minsup a support threshold.

- Through (*frequent*) *closed* and *maximal* itemsets we attempt to limit the output size by reducing redundancy.
- Frequent closed itemsets provide a lossless representation of frequent itemsets and their supports.
- Maximal itemsets attain a more drastic redundancy reduction but provide a lossy representation of frequent itemsets and their supports.

(Frequent) closed itemsets

Definition (Closed Itemset)

An itemset $X \subseteq I$ is *closed* w.r.t. T if for each superset $Y \supset X$ we have $\mathsf{Supp}_T(Y) < \mathsf{Supp}_T(X)$. If $\mathsf{Supp}_T(X) \geq \mathsf{minsup}$, X is *frequent closed*.

In other words: *X is closed if its support decreases as soon as an item is added.*

Notation

- $\mathrm{CLO}_T = \{X \subseteq I \ : \ X \text{ is closed w.r.t } T\}$
(*set of closed itemsets*)
- $\text{CLO-F}_{T,\mathrm{minsup}} = \{X \in \mathrm{CLO}_T \ : \ \mathsf{Supp}_T(X) \geq \mathsf{minsup}\}$
(*set of frequent closed itemsets*)

Obs. Notion of closure does not depend on minsup

Maximal itemsets

Definition (Maximal Itemset)

An itemset $X \subseteq I$ is *maximal* w.r.t. T and minsup if $\mathsf{Supp}_T(X) \geq$ minsup and for each superset $Y \supset X$ we have $\mathsf{Supp}_T(Y) <$ minsup.

In other words: X *is maximal if it is frequent and becomes infrequent as soon as an item is added.*

Notation

- $\mathrm{MAX}_{T,\mathrm{minsup}} = \{X \subseteq I \,:\, X \text{ is maximal w.r.t. } T \text{ and minsup}\}$ (*maximal itemsets*)

When clear from the context, the subscripts will be omitted

Exercise

Dataset T	
TID	ITEMs
1	ABC
2	ABCD
3	BCE
4	ACDE
5	DE

For minsup = $2/5$, identify:

(a) A maximal itemset

(b) A frequent closed itemset which is not maximal

(c) A closed itemset which is not frequent

(a) ABC (2/5) $\in$ MAX

(b) AC (3/5) $\in$ CLO-F \ MAX

(c) ACDE (1/5) $\in$ CLO \ F

BCE (1/5) $\in$ "

Exercise

Dataset T	
TID	ITEMs
1	ABC
2	ABCD
3	BCE
4	ACDE
5	DE

For minsup = 2/5, identify:

(d) Set F (frequent itemsets)

(e) Set CLO-F

(f) Set MAX

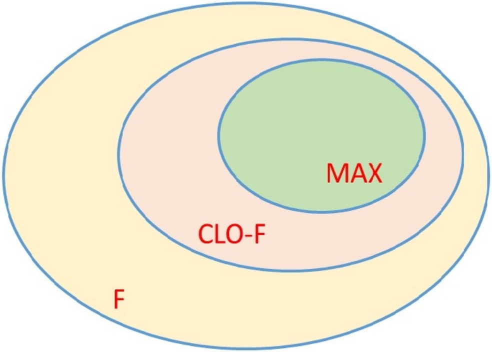
F: A(3) B(3) C(4) D(3) E(3) AB(2) AC(3) AD(2)
BC(3) CD(2) CE(2) DE(2) ABC(2) ACD(2) 14

CLO_F: C, D, E, AC, BC, CE, DE, ABC, ACD 9

MAX: CE, DE, ABC, ACD 4

Properties of closed/maximal itemsets

Property 1: MAX $\subseteq$ CLO-F $\subseteq$ F.



Properties of closed/maximal itemsets

Property 2: for each itemset X there is a closed itemset X' such that

- $X \subseteq X'$
- $\text{Supp}_T(X') = \text{Supp}_T(X)$

X' can be regarded as a "representative" for X

For example:

Dataset T	
TID	ITEMs
1	ABC
2	ABCD
3	BCE
4	ACDE
5	DE

$X = AE\ (1|5)$ $X' = ACDE\ (1|4)$

$X = AD\ (2|5)$ $X' = ACD\ (2|5)$

$X = C\ (4|5)$ $X' = C\ (4|5)$

Properties of closed/maximal itemsets

Property 3: for each frequent itemset X there is a maximal itemset X' such that

- $X \subseteq X'$

X' can also be regarded as a "representative" for X, but it provides NO INFO on SUPPORT!

For example:

Dataset T	
TID	ITEMs
1	ABC
2	ABCD
3	BCE
4	ACDE
5	DE

$X = AC \ (3/5)$

$X' = ABC \ (2/5)$

$minsup = 2/5$

Properties of closed/maximal itemsets

Property 2 : How can we construct X' from X (X' is closed $\wedge$ $Supp_T(X') = Supp_T(X)$)

Given X add items to it as long as the support does not change

When no more items can be added the resulting itemset X' is closed and $Supp_T(X) = Supp_T(X')$

Properties of closed/maximal itemsets

Property 3: How to construct a maximal itemset X' containing a given frequent itemset X?

Same approach as before (exercise)

Consequences of the properties

Can we determine F from:

- CLO-F?
- MAX?

YES

Can we determine F with supports from:

- CLO-F with supports? YES
- MAX with supports? NO

Consequences of the properties

* How do we determine F from Clo_F, MAX ?

By Properties 2 and 3 we know that for each $X \in F$

- There is $X' \in Clo\_F$ such that $X \subseteq X'$
- There is $X' \in MAX$ such that $X \subseteq X'$

$\Rightarrow$ Derive F by taking all subsets of each itemset in Clo_F, or all subsets of each itemset in MAX (they are all frequent!)

Consequences of the properties

We can generate F (+supports) from CLO_F (+supports) as follows (not possible from MAX(+supports))

- Generate F from CLO_F as above
- For each $X \in F$ set

$$Supp_T(X) = \max\{Supp_T(Y) : Y \supseteq X \wedge Y \in CLO\_F\}$$

Justification: Among all $Y \in CLO\_F$ and such that $Y \supseteq X$ (hence they have supports all $\leq Supp_T(X)$) there must be one with exactly the same support as X (by Property 2)

Representativity of closed itemsets

For $X \subseteq I$, let T_X denote the set of transactions where X occurs.

Definition (Closure)

$$\text{Closure}(X) = \bigcap_{t \in T_X} t.$$

Example:

Dataset T	
TID	ITEMs
1	ABC
2	ABCD
3	BCE
4	ACDE
5	DE

$X = AB$ Closure$(X) = ABC$

Closure(X) provides a richer info than X about $T_X \Rightarrow$ Closure(X) can be regarded as a good representative for X

$X = BC$
Closure$(X) = BC$

Representativity of closed itemsets

Properties of closure

Let $X \subseteq I$. If $\mathsf{Supp}_T(X) > 0$ we have:

1. $X \subseteq \mathrm{Closure}(X)$
2. $\mathsf{Supp}_T(\mathrm{Closure}(X)) = \mathsf{Supp}_T(X)$
3. $\mathrm{Closure}(X)$ is closed.

1 Trivial since $X \subseteq t \; \forall t \in T_X$ hence,

$$X \subseteq \bigcap_{t \in T_X} t = \mathrm{Closure}(X)$$

2 Observe that

Representativity of closed itemsets

* X, Closure(X) are contained in every transaction of T_X
* No transaction outside T_X can contain X hence Closure(X)

$\Rightarrow$ T_X is the set of all transactions that contain X and the set of all transactions that contain Closure(X)

$$\Rightarrow \quad \mathrm{Supp}_T(X) = \mathrm{Supp}_T(\mathrm{Closure}(X))$$

Representativity of closed itemsets

③ Is Closure(X) closed ?

If we add one item a to Closure(X) the support must decrease since otherwise a would occur in all transactions that contain Closure(X) ⇒ a would occur in all transactions of T_X ⇒ a would be in their intersection which is Closure(X), by definition ⇒ contradiction

Observations

- Each closed itemset Y represents compactly all (possibly many) itemsets X such that $\mathrm{Closure}(X) = Y$.
- There exist efficient algorithms for mining maximal or frequent closed itemsets.
- Notions of closure similar to the ones used for itemsets are employed in other mining contexts

  (e.g., *graph mining, DNA sequence analysis*).

QUESTION: Does restricting the attention to maximal/frequent closed itemsets avoid the output explosion?

ANSWER:

- **In practice:** they yield a quite effective reduction of the output size;
- **However:** there are non-trivial pathological cases where their number is exponential in the input size (see exercise next)

Exponentiality of maximal/frequent closed itemsets

Exercise

Consider a dataset $T = \{t_1, t_2, \ldots, t_{2n}\}$ be a set of $2n$ transactions, over $I = \{1, 2, \ldots, 2n\}$, for some $n > 0$, where $t_j = I - \{j\}$. Fix minsup=0.5.

1. For each intemset X determine $\mathrm{Supp}_T(X)$ as a function of n and $|X|$.
2. Which are the *maximal itemsets*?
3. Show that **there is an exponential number of maximal itemsets**. (Use the fact that $\binom{a}{b} \geq (a/b)^b$)
4. Show that **there is an exponential number of frequent closed itemsets**.

Solution to the exercise

Let's view T as a matrix:

$n=4$

Transactions/items	1	2	3	4	5	6	7	8
t_1	0	1	1	1	1	1	1	1
t_2	1	0	1	1	1	1	1	1
t_3	1	1	0	1	1	1	1	1
t_4	1	1	1	0	1	1	1	1
t_5	1	1	1	1	0	1	1	1
t_6	1	1	1	1	1	0	1	1
t_7	1	1	1	1	1	1	0	1
t_8	1	1	1	1	1	1	1	0

Itemset $X \subseteq t_i \iff i \notin X$

E.g. $X = \{2,5,6,7\}$ contained in t_1, t_3, t_4, t_8

Solution to the exercise

① $\mathrm{Supp}_T(X) = ?$

$$\mathrm{Supp}_T(X) = \frac{2n - |X|}{2n} = 1 - \frac{|X|}{2n}$$

② W.r.t. minsup = 0.5 what are the maximal itemsets?

Maximal itemsets are those with $|X| = n$

$|X| > n \Rightarrow$ not frequent

$|X| \leq n \Rightarrow$ frequent

$|X| = n \Rightarrow$ maximal

Solution to the exercise

③ $|MAX| = ?$

$$|MAX| = \binom{2n}{n} \geq (2n/n)^n = 2^n$$

$$|input| \simeq (2n)^2$$

$$\Rightarrow \quad |MAX| = \Theta\left(2^{\sqrt{|input|}}\right)$$

④ $|CLO\_F| = ?$ $\qquad MAX \subseteq CLO\_F$

$|CLO\_F| \geq |MAX| \Rightarrow$

$|CLO\_F|$ exponential in $|input|$

Solution to the exercise

How about if we impose explicitly a limit on the output size?

Top-K frequent itemsets

Let T be a dataset of N transactions over a set I of d items.

Definition (Top-K frequent itemsets)

Let $X_1, X_2, \ldots,$ an enumeration of the non-empty itemsets in non-increasing order of support (ties broken arbitrarily). For a given integer $K \geq 1$. let

$$s(K) = \mathsf{Supp}_T(X_K).$$

The Top-K frequent itemsets w.r.t. T are all itemsets of support $\geq s(K)$.

Observations:

- The number of Top-K frequent itemsets is $\geq K$.
- Why not considering only $X_1, \ldots, X_K$?
- $s(K)$ can be defined *equivalently* as the maximum support threshold that provides at least K frequent itemsets.

Example

DATASET T	
TID	**ITEMS**
1	ABC
2	ABCD
3	BCE
4	ACDE
5	DE

Top-K Frequent Itemsets

Support = 4/5

Support = 3/5

Support = 2/5

Itemsets: C, A, B, D, E, AC, BC, AB, AD, CE, DE, ABC, ACD, …

Top-1 frequent itemsets: C

Top-K frequent itemsets (K=2,..7): C,A,B,D,E,AC,BC

Top-K frequent itemsets (K=8,..13): C,A,B,D,E,AC,BC,AB,AD,CE,DE,ABC,ACD

Top-K frequent closed itemsets

Definition (Top-K frequent closed itemsets)

Let $X_1, X_2, \ldots,$ an enumeration of the non-empty closed itemsets in non-increasing order of support (ties broken arbitrarily). For a given integer $K \geq 1$. let

$$sc(K) = \text{Supp}_T(X_K).$$

The Top-K frequent closed itemsets w.r.t. T are all closed itemsets of support $\geq sc(K)$.

The same observations made for the Top-K frequent itemsets hold for the Top-K frequent closed itemsets.

Example

DATASET T	
TID	**ITEMS**
1	ABC
2	ABCD
3	BCE
4	ACDE
5	DE

Top-K
Frequent Closed Itemsets

Support = 4/5

Support = 3/5

Support = 2/5

Itemsets: C, D, E, AC, BC, CE, DE, ABC, ACD, ...

Top-1 frequent closed itemsets: C

Top-K frequent closed itemsets (K=2,..5): C,D,E,AC,BC

Top-K frequent closed itemsets (K=6,..9): C,D,E,AC,BC,CE,DE,ABC,ACD

Observations

- There are efficient algorithms for mining the Top-K frequent (closed) itemsets. A popular strategy is this:
  - Generate (closed) itemsets in non-increasing order of support (using a priority queue).
  - Stop at the first itemset smaller support than the K-th one.
- The notion of Top-K patterns (w.r.t. some ranking) is used a lot in many different contexts.

Obs. Top-K frequent (closed) itemsets are the output of a STANDARD frequent (closed) itemset problem with a support threshold that depends on K

QUESTION: Does the "Top-K" formulation avoid the output explosion?

How well does K relates to the output size?

ANSWER:

- **In practice:** yes!
- **However:**
  - For Top-K frequent closed itemsets there is a tight polynomial bound on the output size;
  - For Top-K frequent itemsets there are pathological cases with exponential output size.
  - For some applications, setting the support threshold explicitly may be *more natural*;

Control on the output size

The next theorem (without proof) shows that for Top-K frequent closed itemsets, K provides tight control on the output size.

Theorem

For $K > 0$, the Top-K frequent closed itemsets are $O(d \cdot K)$, where d is the number of items.

Hence: for small K (at most polynomial in the input size) the number of Top-K frequent closed itemsets will be polynomial in the input size.

Control on the output size

The next exercise shows that, unfortunately, exponentiality of the output size can occur when lifting the restriction to closed itemsets.

Exercise

Let d be an even integer, and define T as the set of the following $N = (3/2)d$ transactions over $I = \{1, 2, \ldots, d\}$

$$
\begin{aligned}
t_i &= \{i\} \quad 1 \le i \le d \\
t_{d+i} &= I - \{i\} \quad 1 \le i \le d/2.
\end{aligned}
$$

1. Identify the itemsets of support $= 1/3$ and those of support $> 1/3$.
2. Using the result of the previous point, show that the number of Top-K frequent itemsets, with $K = d$, is exponential in d.

To recap ...

Maximal itemsets:

- Lossless representation of the frequent itemsets (lossy, if supports are required);
- In practice, much fewer than the frequent itemsets, but, in pathological cases, still exponential in the input size.

$\Rightarrow$ reduction of redundancy

Frequent closed itemsets:

- Lossless representation of the frequent itemsets with supports;
- In practice, much fewer than the frequent itemsets, but, in pathological cases, still exponential in the input size.

$\Rightarrow$ reduction of redundancy

Top-K frequent (closed) itemsets:

- Output size: $O(d \cdot K)$, if restricted to closed itemsets, otherwise small in practice but exponential in d for pathological cases.

$\Rightarrow$ reduction of redundancy and control on the output size (with closed itemsets)

Exercises

Exercise

Let T be a dataset of transactions over I. Let $X, Y \subseteq I$ be two closed itemsets and define $Z = X \cap Y$.

1. Find a relation among T_X, T_Y and T_Z (i.e., the sets of transactions containing X, Y, and Z, respectively). Justify your answer.
2. Show that Z is also closed.

Exercise

Let $I = \{a_1, a_2, \ldots, a_n\} \cup \{b_1, b_2, \ldots, b_n\}$ be a set of $2n$ item, e let $T = \{t_1, t_2, \ldots, t_n\}$ be a set of n transactions over I, where

$$t_i = \{a_1, a_2, \ldots a_n, b_i\} \qquad \text{per } 1 \leq i \leq n.$$

For $\text{minsup} = 1/n$, determine the number of frequent closed itemsets and the number of maximal itemsets.

Exercises

Exercise

Let T be a dataset of N transactions over a set of items I, and let $\epsilon > 0$ be a parameter. For each itemset X, let s_X be an approximation of its true support $\mathrm{Supp}_T(X)$ such that

$$\mathrm{Supp}_T(X) - \epsilon \leq s_X \leq \mathrm{Supp}_T(X) + \epsilon$$

Consider an ordering of all itemsets $X_1, X_2, X_3, \ldots$ by non-increasing approximate support ($s_{X_1} \geq s_{X_2} \geq s_{X_3} \ldots$), and let $K < K'$ be two indices such that $s_{X_K} > s_{X_{K'}} + 2\epsilon$.

1. Show that if $i \leq K$ and $j \geq K'$, then $\mathrm{Supp}_T(X_i) > \mathrm{Supp}_T(X_j)$
2. Based on the previous points, show that the set $\{X_1, X_2, \ldots, X_{K'}\}$ contains the Top-K frequent itemsets with respect to the true support.

References

TSK06 P.N.Tan, M.Steinbach, V.Kumar. Introduction to Data Mining. Addison Wesley, 2006. Chapter 6 (Section 6.4).

LRU14 J.Leskovec, A.Rajaraman and J.Ullman. Mining Massive Datasets. Cambridge University Press, 2014. Chapter 6